Development of a Piezoelectric Energy Regeneration System for Roads and Pathways

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1.1 Introduction to Problem

Global Energy Sources

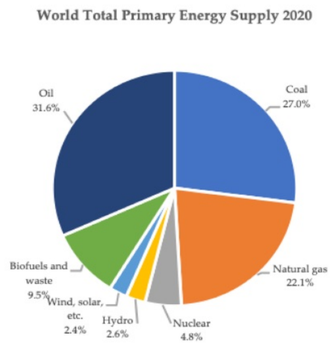


Fig. 1.1.1 Global Energy Sources[9]

Rising Electricity demands

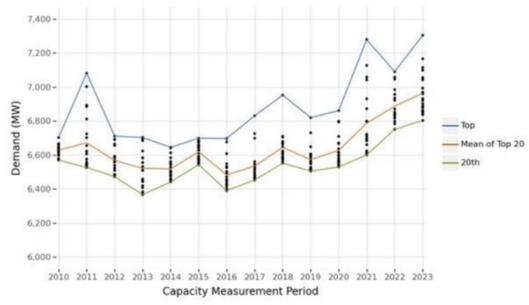


Fig. 1.1.2 Top 20 largest peak demand periods from 2010-2023, with the top, mean and 20th positions highlighted

76.7% from fossil fuels → emit a lot of greenhouse gases → accelerate climate change & threaten public health

1.2 The Piezoelectric Effect

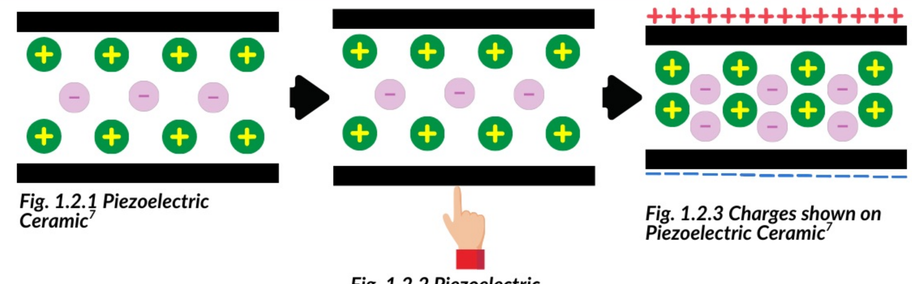


Fig. 1.2.1 Piezoelectric Ceramic[7]

Fig. 1.2.2 Piezoelectric Ceramic being squeezed[7]

Fig. 1.2.3 Charges shown on Piezoelectric Ceramic[7]

- In Piezoelectric Ceramics, the charges are placed in an asymmetrical manner
- Despite this, the charges are still balanced and no net charge is found on the faces
- When the ceramic is compressed, the charges do not balance, and there are positive and negative charges on the opposite faces
- This causes an AC voltage spike to be produced (Due to rebounce)

1.3 Development Goal

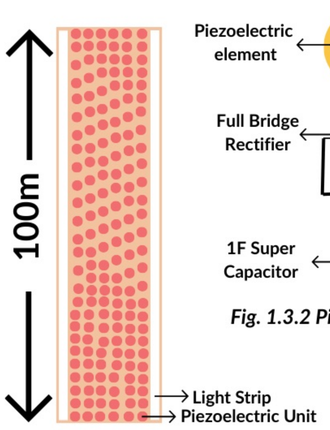


*Fig. 1.3.1 Pathway *Upscaled*[7]

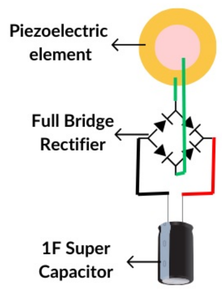


Fig. 1.3.2 Piezoelectric Unit[7]

- Starting off with a 100 m pathway, an array of Piezoelectric Units will be placed.
- Each unit consists of 1 Piezoelectric ceramic, 1 full-bridge rectifier, and a capacitor. All the capacitors will be interconnected to form 1 large array
- When someone steps on the elements, an AC Voltage Spike is produced, which is converted to DC through the rectifier.
- This energy is then stored in the capacitor to power the light strip at night

2. Methodology

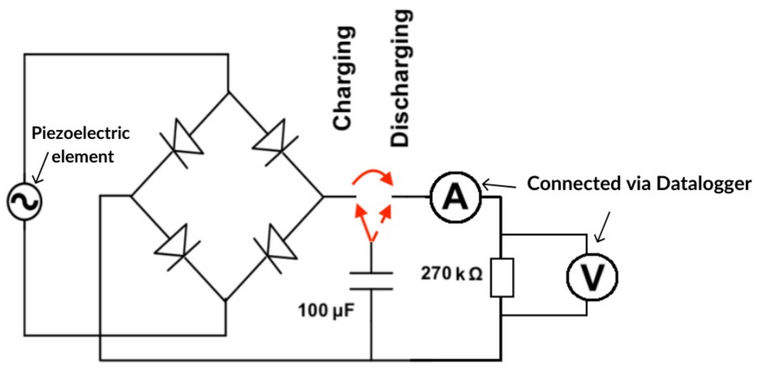


Fig. 2.1 Experimental Setup Diagram[7]

1. Strike the piezoelectric ceramic 50x while the switch is in the "Charging mode"
2. Flip the switch and record the spike readings on the Ammeter and the Voltmeter using a datalogger and LoggerPro application
3. Repeat the experiment and find out the average results
4. *The force results were measured using an iOLab unit, separate to this setup

3. Results and Key Findings

1 *Power developed by 50 strikes of 10N force*

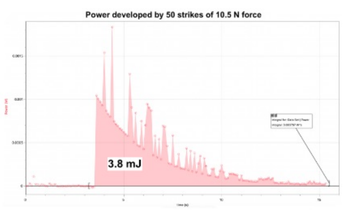


Fig. 3.1 Power developed from 50 strikes of 10N[7]

2 *Power developed by 50 strikes of 10.5N force*

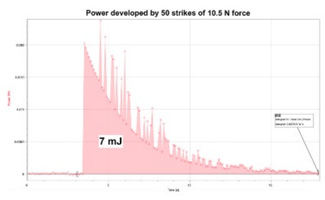


Fig. 3.2 Power developed from 50 striked of 10.5N[7]

4 *Voltage output from capacitor after 50 strikes of 10.5N force*

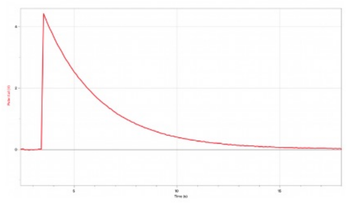


Fig. 3.3 Capacitor voltage output[7]

5 *Current output from capacitor after 50 strikes of 10.5N force*

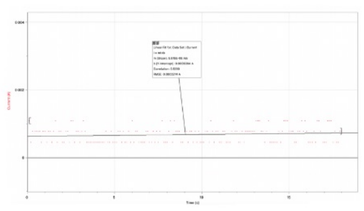


Fig. 3.4 Capacitor current output[7]

3 *Force applied by 1 strike (Average)*

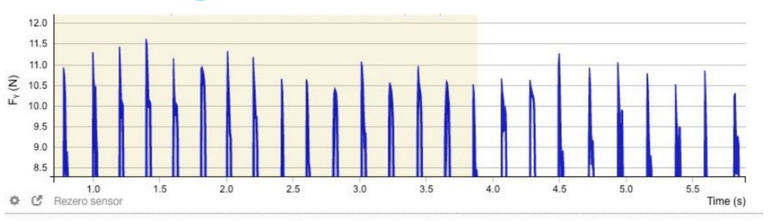


Fig. 3.5 Measurement of force of 1 strike (Averaged out)[7]

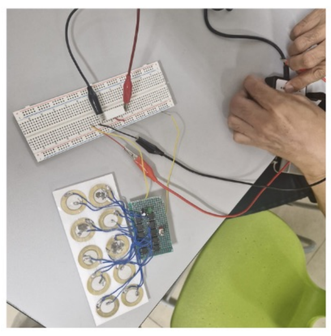


Fig. 3.6 Setting up system[7]

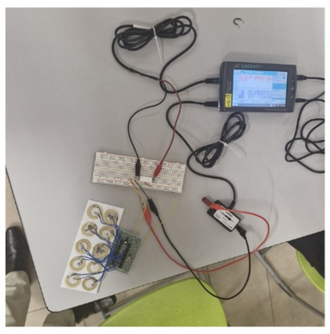


Fig. 3.7 Gathering Data[7]

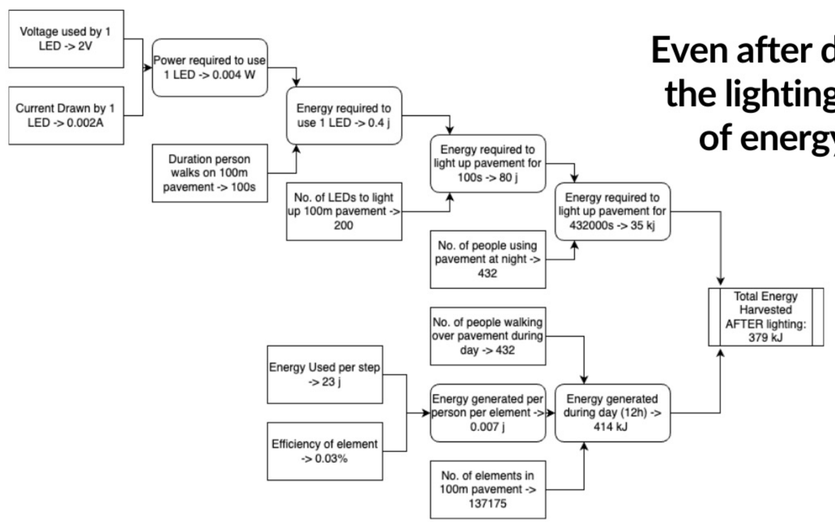
4. Discussion

4.1 Cost for 100 m Application

1. **Cost of Piezoelectric elements for 100m Application:** 7c (cost per piece) X 137175 (Amount needed) = **$9600**
2. **Cost of Electrical components:** (5c (Cost per chip + capacitor) X 137175) + ($50 (Cost of LED strip) X 2) = **$6958**
3. **Misc Cost: $442**

Total Initial Cost: $17000

4.2 Energy Harvesting



Even after deducting the energy used in the lighting system, there is still 379 kJ of energy stored in the capacitors!

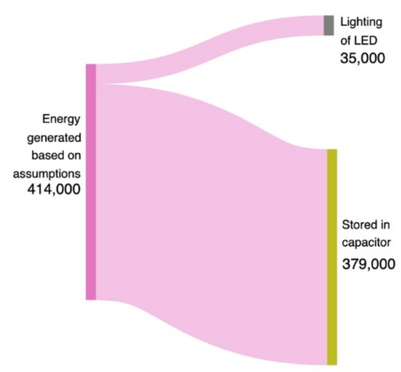


Fig. 4.2.1 Flowchart of energy harvesting system calculations[7]

*This model assumes 432 people walk on the pathway during the day, and 432 people walk on the pathway at night.
*This model does not consider the energy generated by those that walk on the pathway at night.
*This model assumes that all 432 people walk on the 100m pathway for exactly 100s

5. Energy Harvesting Applications

5.1 Power Generation

- Global net-zero goals → increase demand for clean energy solutions to reduce carbon emissions
- Reduces dependence on mains power for street-lamp lighting. Useful for emergency situations

Fig 5.1.1 UNSDG logos[6]

5.2 Resolution

- The system effectively harvests more energy than it consumes, indicating it is energy-positive and potentially useful for sustainable urban lighting or similar applications.
- However, if the efficiency of the piezoelectric element can be brought up from 0.03%, the energy harvested would drastically increase, based on information found in other research papers

Fig 5.2.1 Efficiency symbol[8]

5.3 Applications

Fig 5.3.1 Energy symbol[8]

- Can be deployed on all major pathways at a relatively low cost compared to construction costs
- Can even extend to roads for higher energy harvesting
- Possible to connect to grid if the power generated is sufficient

6. Conclusion & Future work

While the energy generation efficiency of piezoelectric materials remains limited, our research highlights their strong potential as accurate and responsive components in smart sensing applications. This positions them not only as a supplementary energy source, but also as a valuable technology for enhancing sensor-based infrastructure. We have more to explore in the future, such as:

- Exploring the viability of this system on Roads - Activate street lamps for the next 200m ahead, 100m behind when a car passes by.
- Exploring the viability of this system as wearable shoes
- Exploring more efficient methods of kinetic energy conversion
- Add an automatic lighting mechanism to the prototype, such that when someone steps on the piezoelectric element at night, the lights 10m in front and 5m behind automatically light up

1. References
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